Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/30/2011
PI Name:	Mednieks, Maija Ph.D.		
Project Title:	Protein Expression in Salivary Glands: Effects of Extended Space Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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City:	Farmington	**State:**	CT
Zip Code:	06030-0001	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Shuttle		
Flight Assignment:	STS-131 ; STS-133		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hand, Arthur D.D.S. (University of Connecticut Health Center)		
Grant/Contract No.:	NNX09AP13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed study is to determine effects of extended space travel on oral tissues using the salivary glands of mice. Proteins secreted from the salivary glands play an important role in maintaining the integrity of the mouth and teeth. Rodent salivary glands are structurally and functionally similar to those of humans therefore useful for research on changes associated with travel in space. Namely, studying the effects of zero gravity (ØG) on tissue, cellular and molecular levels. Our previous studies on short duration (~ 1 week) space flight missions showed that specific secretory proteins were modified by exposure to both zero and hyper - G conditions. Objective: to test the hypothesis that a long duration (~ 1 month) space flight will show effects of extended ØG on protein expression in salivary glands of the experimental animals. Methods: salivary glands from flight animals and ground-based controls will be prepared for biochemical (electrophoresis and Western and Northern blotting) and morphological (electron microscopy and immunocytochemistry) experiments to determine changes in ultrastructure and in expression and intracellular localization of proteins that are markers for different cell types in the salivary glands or are indicators of specific cell functions. Several of the proteins have previously been shown to respond to a variety of environmental stimuli, including space flight. Significance: proteins present in secretion granules of salivary glands are released into saliva and can be used to monitor effects of environmental stress. Employing saliva as a diagnostic test medium could have clinical applications and be useful in measuring human responses to space flight. Conclusions: measuring and quantification of proteins in salivary glands will show if space travel affects the physiology of the oral cavity as well as the regulation of cellular signaling and molecular signal transduction of secretory cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Background and Objectives: It is important to study the effects of travel in space and to measure reactions to the stress of weightlessness in zero gravity (ØG) conditions. Stress changes hormone action that is reflected in different body parts/organs as well as in glands that produce saliva. The goal is to devise a clinical test for stress disorders in space and on Earth. Saliva contains substances (e.g. proteins) that change during travel in space. We can determine which salivary proteins have been changed by using biochemical tests and histologic analyses to study tissues and cells (salivary glands) that produce saliva in mice flown on the space shuttle Discovery STS-131 and 133 missions. This information we will be used to design a test kit to measure responses in astronauts so no blood will have to be drawn. Collecting saliva instead of blood and urine, will be easier during space flight and will make it possible to study hormone changes experienced by Astronauts and Cosmonauts at ØG. Travel in space is an ideal model system for studying a specific environmental impact on physiologic function that is due to gravitational change and cannot be duplicated on earth. Understanding the effect of ØG will define changes due to the environment as the stressor. This information can then be adapted to devising specific tests of different conditions using saliva for general clinical applications in medicine in space or on Earth. Research Description: The Biospecimen Sharing Program has provided an unparalleled opportunity to study tissue and secretory protein signaling in tissues affected by environmental stimuli whereby cellular and molecular changes associated with travel in space can be identified in experimental animals and can be monitored by measuring the expression of specific proteins. Previous studies on the effects of space flight have shown that regulation by catecholamine hormones is markedly altered during space flight. The expression of specific proteins as markers can potentially be used for measuring responses to environmental stress since changes in the expression of secretory proteins are reflected in saliva.
Task Progress:	Results from STS-131: Protein expression is considerably altered during space flight and shown by electrophoresis, Western Blotting and immunocytochemistry. Two major salivary proteins were affected: one, PKA RII showed a decrease while another (DCPP) showed an increase in the flight animals compared to controls. A cellular enzyme (PDE) was decreased in flight animals. Proteins from heart muscle showed variability in the electrophoretic banding patterns of individual animals as well as differences between flight and controls. The ultrastructure of salivary glands is only somewhat, but not significantly altered during space flight. Preliminary findings from STS-133 show similar results. Hardware. Specialized instruments designed for dissecting small tissue samples were employed. Control group animals were housed in the same environmental modules (AEM) as those on the flight. Tissue collection was successful due to availability of dissecting microscopes provided by the Carl Zeiss Co. Electron micrographs, immunocytochemistry and morphometry were carried out using services and instrumentations of the UConn Health Center Electron Microscope facility. Reports • STS-133 30 Day Post Flight Operational Report was submitted and is summarized below. Methods employed were identical with those used for STS-131. The extended recovery protocol of this mission is extremely important in determining if the changes that were observed return to normal values. No anatomical anomalies of either salivary gland or heart ventricle were observed at the time of dissection. Initial/preliminary experiment of D+1 suggests that specific secretory proteins are altered.
Bibliography Type:	Description: (Last Updated: 02/04/2021)
Abstracts for Journals and Proceedings	Mednieks M, Hand A. "Protein Expression in Heart and Salivary Glands: Effects of Extended Space Flight." The 44th Annual Conference on Gene Families and Isozymes, Providenciales in the Turks & Caicos Islands of the British West Indies, December 5-10, 2010. The 44th Annual Conference on Gene Families and Isozymes, Providenciales in the Turks & Caicos Islands of the British West Indies, December 5-10, 2010. , Dec-2010
Abstracts for Journals and Proceedings	Mednieks M, TsesisM, Manz D, Larson N, Hand AR. "Protein Expression in Mouse Salivary Glands: Effects of Spaceflight." IADR/AADR/CADR 89th General Session and Exhibition: Saliva and Salivary Glands in Health and Disease, San Diego, CA, March 16-19, 2011. Part 1, Poster Discussion Session, Session Chairs: S. HU and M. MEDNIEKS. Journal of Dental Research 2011;90:2372. http://iadr.confex.com/iadr/2011sandiego/webprogram/Paper148357.html , Mar-2011

Abstracts for Journals and Proceedings	Mednieks M, Hand A. "Protein expression in Salivary glands After Space Flight." Spring Symposium, New England Society for Electron Microscopy, Woods Hole, MA, 6 May 2011. Spring Symposium, New England Society for Electron Microscopy, Woods Hole, MA, 6 May 2011. , May-2011
Abstracts for Journals and Proceedings	Mednieks MI, Tsesis M, Larson N, Hand AR. "STS-131: Protein Expression in Heart and Salivary Glands: Effects of Extended Space Flight." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Piras M, Hand AR, Mednieks MI, Piludu M. "Amylase and cyclic AMP receptor protein expression in human diabetic parotid glands." J Oral Pathol Med. 2010 Oct;39(9):715-21. PMID: 20456617 ; http://dx.doi.org/10.1111/j.1600-0714.2010.00898.x , Oct-2010
Awards	Mednieks M, Hand A. "Poster Presentation Award for: 'Protein expression in Salivary glands After Space Flight.' Spring Symposium, New England Society for Electron Microscopy, Woods Hole, MA, 6 May 2011." May-2011